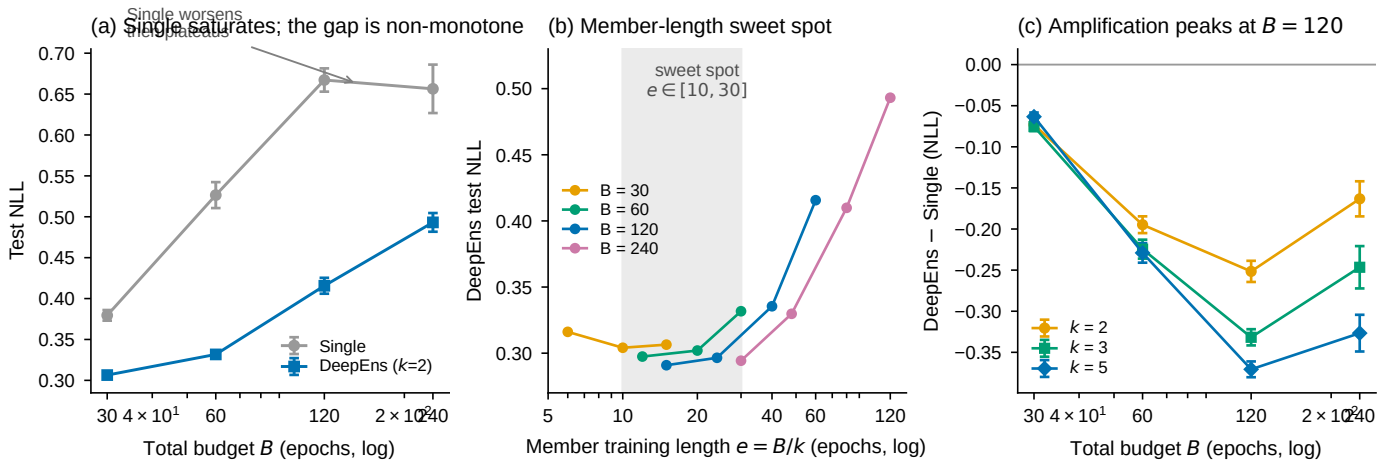
(a) Single saturates; the gap is non-monotone
Single worsens
then plateaus
Test NLL
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
Single
DeepEns (k=2)
30
4 × 10^1
60
120
2 × 10^2
240
Total budget B (epochs, log)
(b) Member-length sweet spot
sweet spot
e ∈ [10, 30]
DeepEns test NLL
0.50
0.45
0.40
0.35
0.30
B = 30
B = 60
B = 120
B = 240
5
10
20
40
60
120
Member training length e = B/k (epochs, log)
(c) Amplification peaks at B = 120
DeepEns − Single (NLL)
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
k = 2
k = 3
k = 5
30
4 × 10^1
60
120
2 × 10^2
240
Total budget B (epochs, log)